OUT-01 Outreach Demonstration Plan

Solar Car for Renewable Energy Demonstration · Portland High School, Mitchells Plain · Ethics ref. EBE REC EBE/03373/2026

Overview

Item	Detail
Audience	About 40 Grade 11 learners
Venue	School hall (indoors); driving route marked with cones
Duration	10–15 min presentation, then a 45–60 min driving demonstration
Team	Jonathan Tirkaso (organiser), A/Prof. David Oyedokun (supervisor), UCT Racing members
Equipment	Solar car, Jonathan's phone (driver controller), iPad on a stand (live dashboard for the audience), laptop and projector, cones and boxes, questionnaires, pens, USB-C charger

Timeline

Time	Activity	Who
−30 min	Set up cone route, iPad on stand, deck on projector; test drive in manual and co-pilot	Jonathan, UCT Racing
0–5 min	Hand out questionnaires; learners complete Side 1 only and keep the sheet	UCT Racing / teacher
5–20 min	Presentation (16 slides)	Jonathan
20–23 min	Demonstration drive: one run in manual, one in co-pilot, explaining the dashboard on the iPad	Jonathan
23–58 min	Learner runs: each learner drives two short runs, one per mode	All
58–65 min	Learners complete Side 2 immediately, before any other presenter; collect all sheets	UCT Racing / teacher

Throughput: 40 learners × (2 runs × ~25 s + ~20 s handover) ≈ 47 min. If running late, shorten the route by moving cones; do not drop a mode, since every driver needs both to answer item D7.

Running the learner drives

- **Queue:** four learners wait at the start line; a UCT Racing member manages the queue and resets cones.
- **Mode order:** learners with an odd Form ID start in manual; even Form IDs start in co-pilot. This balances any practice effect between the two modes.
- **Coaching:** Jonathan stands with each driver, shows the throttle, steering and co-pilot switch, then steps back.
- **Neutral wording:** "Try both modes and decide for yourself." Do not suggest that co-pilot is easier.
- **Audience:** everyone else watches the car and the live dashboard on the iPad.

Route

About 6–8 m long, 20–30 s per run: start → slalom of 3–4 cones → short corridor of boxes (where co-pilot side correction matters most) → box wall at the finish, so drivers see co-pilot stop the car before it hits.

Roles

Person	Responsibilities
Jonathan	Presentation, demonstration drive, coaching each driver
Supervisor	iPad dashboard and emergency STOP; overall supervision
UCT Racing	Questionnaires out and in, queue management, resetting cones

Person	Responsibilities
Teacher (if available)	Learner discipline and order

Safety and data protection

- **Emergency stop:** the iPad also sends keep-alive messages, so the car's 500 ms link-loss stop will not trigger if only the phone disconnects. The supervisor presses STOP on the iPad, or anyone flips the car's power switch, if the car does not respond.
- **Wi-Fi:** cover the network name and password label on the car so learners cannot connect their own devices.
- **Photos and video:** only of learners whose parent and learner both answered Yes on the consent form; keep others out of frame.
- **Questionnaires:** anonymous (Form ID only); learners who ticked No in Section A stop and hand the sheet back.

Checklist: night before

- ☐ Charge the battery fully; check SOC on the dashboard
- ☐ Test driving with the phone and iPad connected at the same time; confirm STOP works from the iPad
- ☐ Test co-pilot against cones and boxes; use boxes where cones are not detected
- ☐ Cover the Wi-Fi label on the car
- ☐ Print about 45 double-sided questionnaires (flip on long edge); write Form IDs 1–45
- ☐ Copy the deck and both videos to a USB stick; test video playback on the presentation laptop
- ☐ Pack: car, phone, iPad and stand, USB-C charger, cones, boxes, pens, signed consent forms list

Checklist: after the event

- ☐ Collect all questionnaires; count returned sheets and note any No consents
- ☐ Store sheets securely; enter responses into the Excel workbook (Form ID only, no names)
- ☐ Remove or blur media of learners without photo consent
- ☐ Do not upload learner data to any generative AI tool; destroy sheets once the report has been assessed
- ☐ Note observations for the report: runs completed, any failsafe events, co-pilot behaviour, timing